



# Beyond Naive Parameter Sharing : A Disentangled-Head MARL Architecture for Bandwidth Constrained Vehicular OTA Updates

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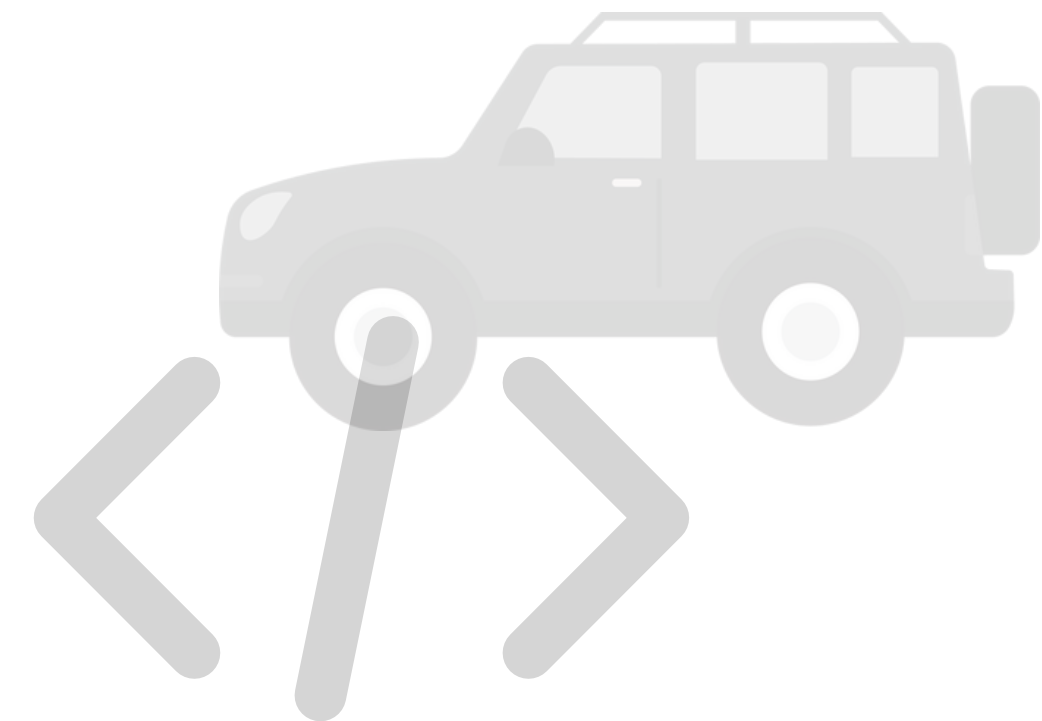
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# INTRODUCTION

## The Scheduling Problem

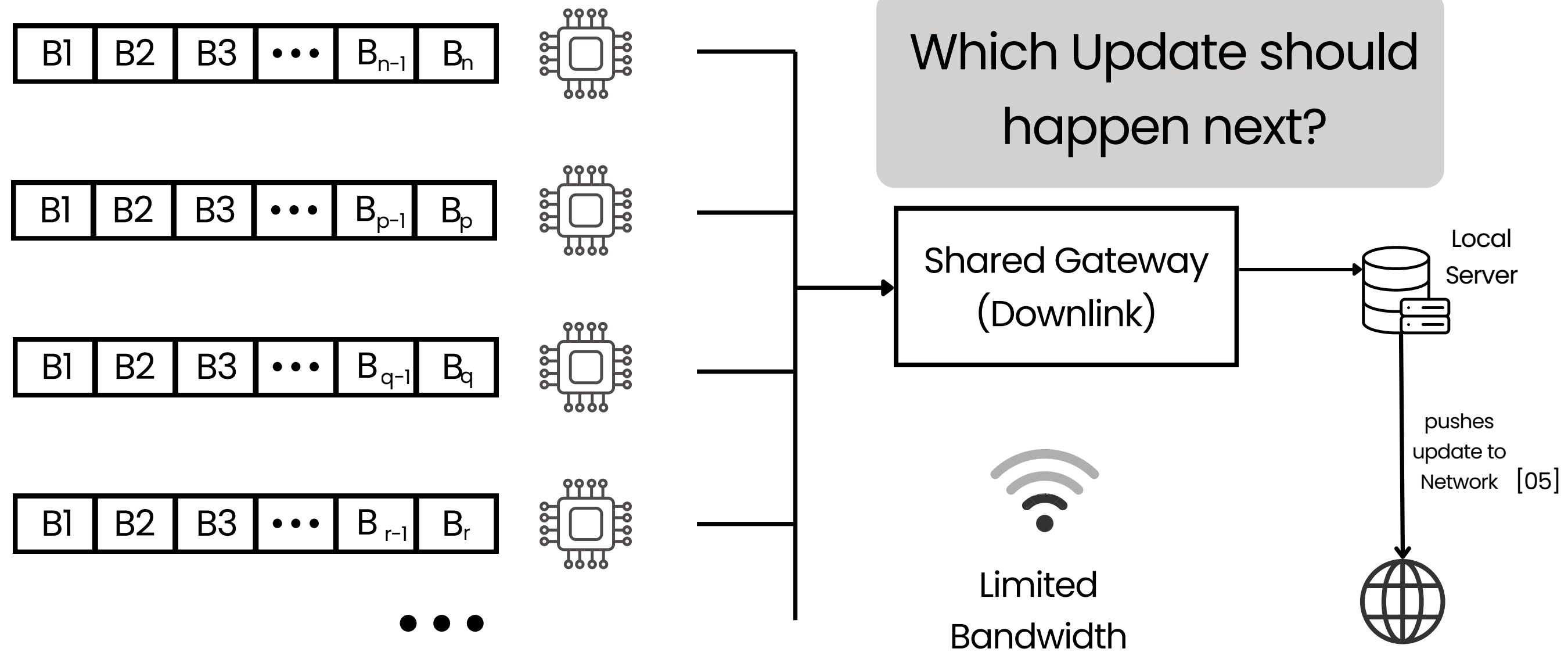


Figure: Heterogeneous ECUs<sup>[01], [05]</sup> competing for shared OTA gateway bandwidth



## RELATED WORKS : IPPO

### Independent Proximal Policy Optimization for Cooperative MARL

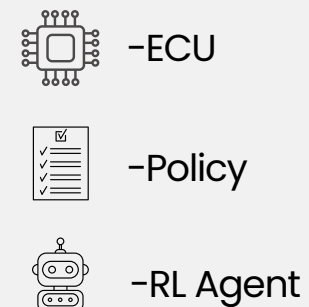
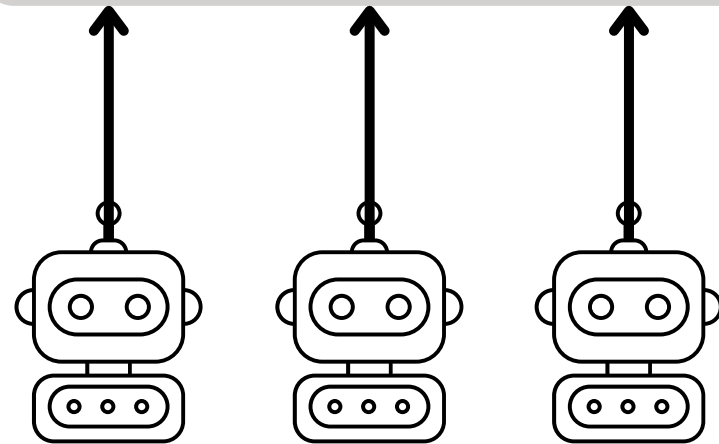
Paper : *Is Independent Learning All You Need in the StarCraft Multi-Agent Challenge?* [04]

Authors: Christian Schröder de Witt et al.

Year: 2020

Venue: NeurIPS 2020

Shared Actor



- Independent policy optimization
- Local observation → local action
- Shared actor parameters

Can simple shared-policy learning coordinate heterogeneous ECUs under a coupled OTA channel?



## RELATED WORKS : MAPPO

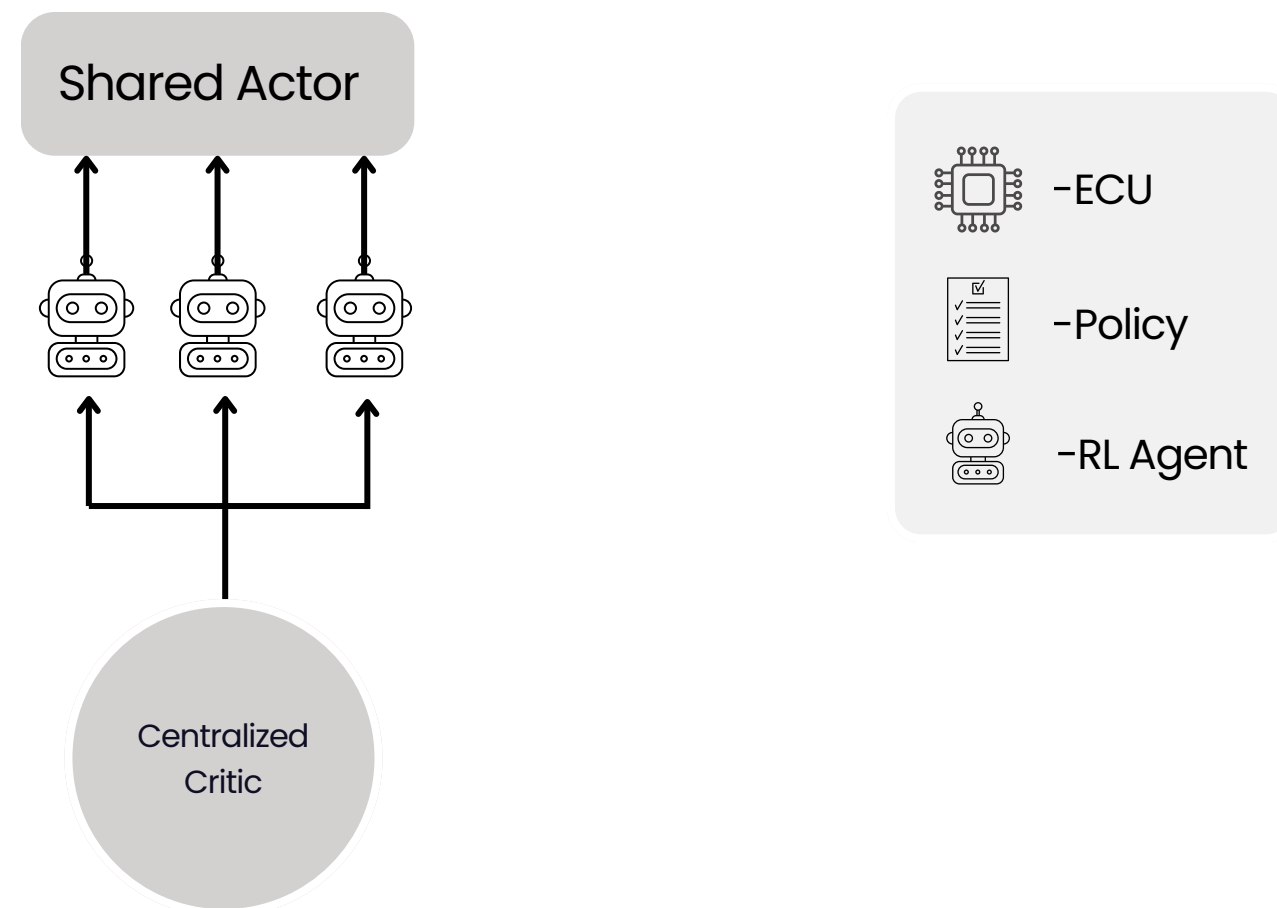
### Centralized Training, Decentralized Execution

Paper : *The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games* [03]

Authors: Chao Yu et al.

Year: 2022

Venue: NeurIPS 2022 — Datasets and Benchmarks Track



- Shared actor
- Centralized critic during training
- Decentralized execution
- Global information helps the critic evaluate cooperative behavior

Does centralized training overcome the coordination difficulty of heterogeneous OTA scheduling?



# RELATED WORKS : FP3O

## Parameter-Sharing Versatility

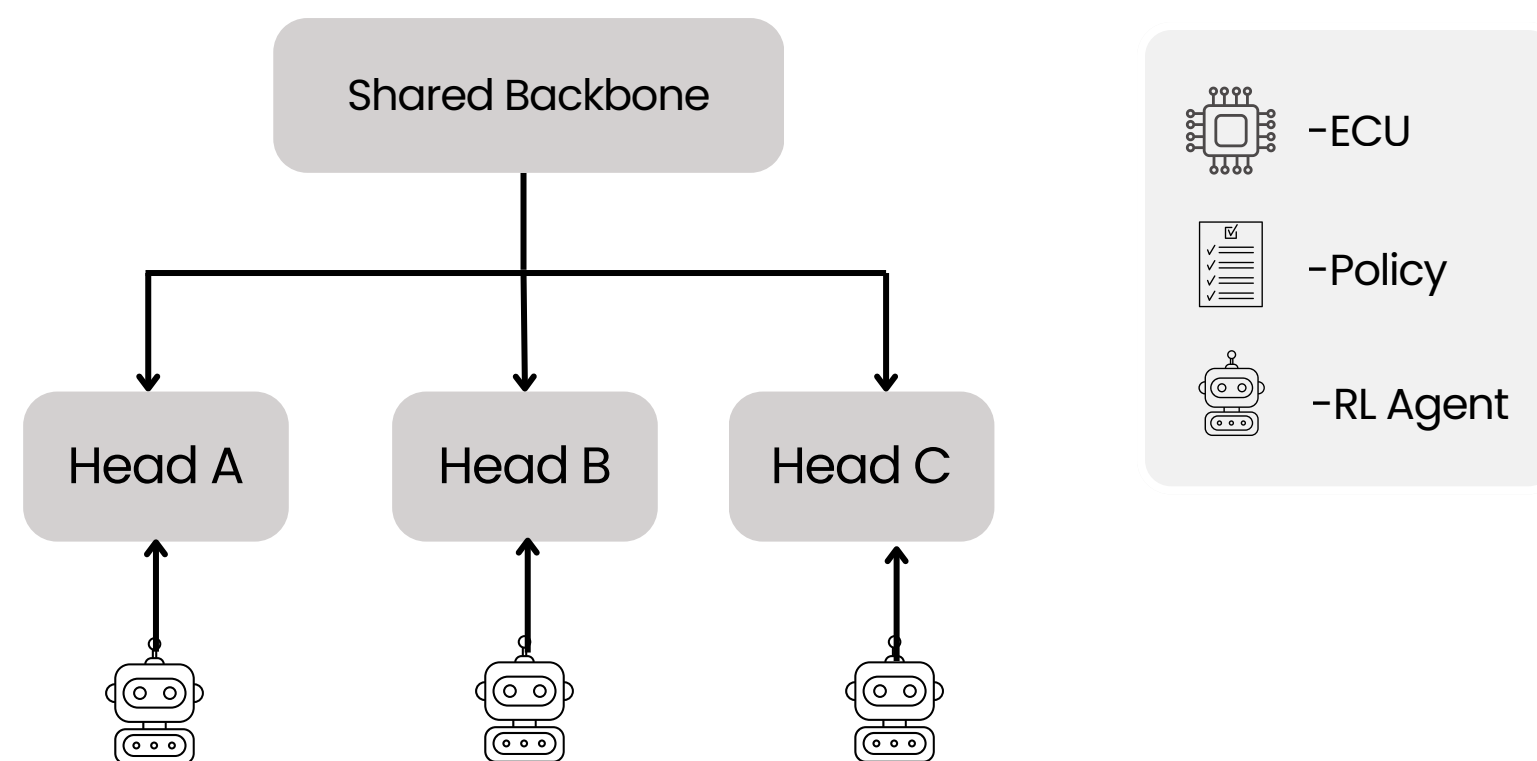
Paper : FP3O : Enabling Proximal Policy Optimization in Multi-Agent Cooperation with Parameter-Sharing Versatility [02]

Authors: Lang Feng, Dong Xing, Junru Zhang, Gang Pan

Year: 2023

Venue: arXiv preprint

Journal publication: IEEE Transactions on Neural Networks and Learning Systems, 2026, online ahead of print.



- Shared representation
- Specialized policy heads
- Allows parameter specialization without completely independent networks

Can parameter specialization preserve shared knowledge while handling heterogeneous ECU roles?



# ◆ RESEARCH OBJECTIVES

| Research Gaps                                                                                                                                        | Research Questions                                                                             | Research Objectives                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| G1 - Heterogeneous policy specialization<br>Parameter sharing dominates MARL, but specialization under heterogeneous ECU roles remains unclear [06]. | RQ1 : Does policy specialization improve MARL under heterogeneous roles and shared contention? | RO1: Develop and evaluate a specialized-head MARL architecture for automotive OTA scheduling. |
| G2 - It is unclear whether type information alone can handle heterogeneous agent roles [07], [08]                                                    | RQ2: How does ECU-type conditioning interact with shared and specialized policies?             | RO2: Compare blind and type-conditioned IPPO, MAPPO, and FP3O                                 |



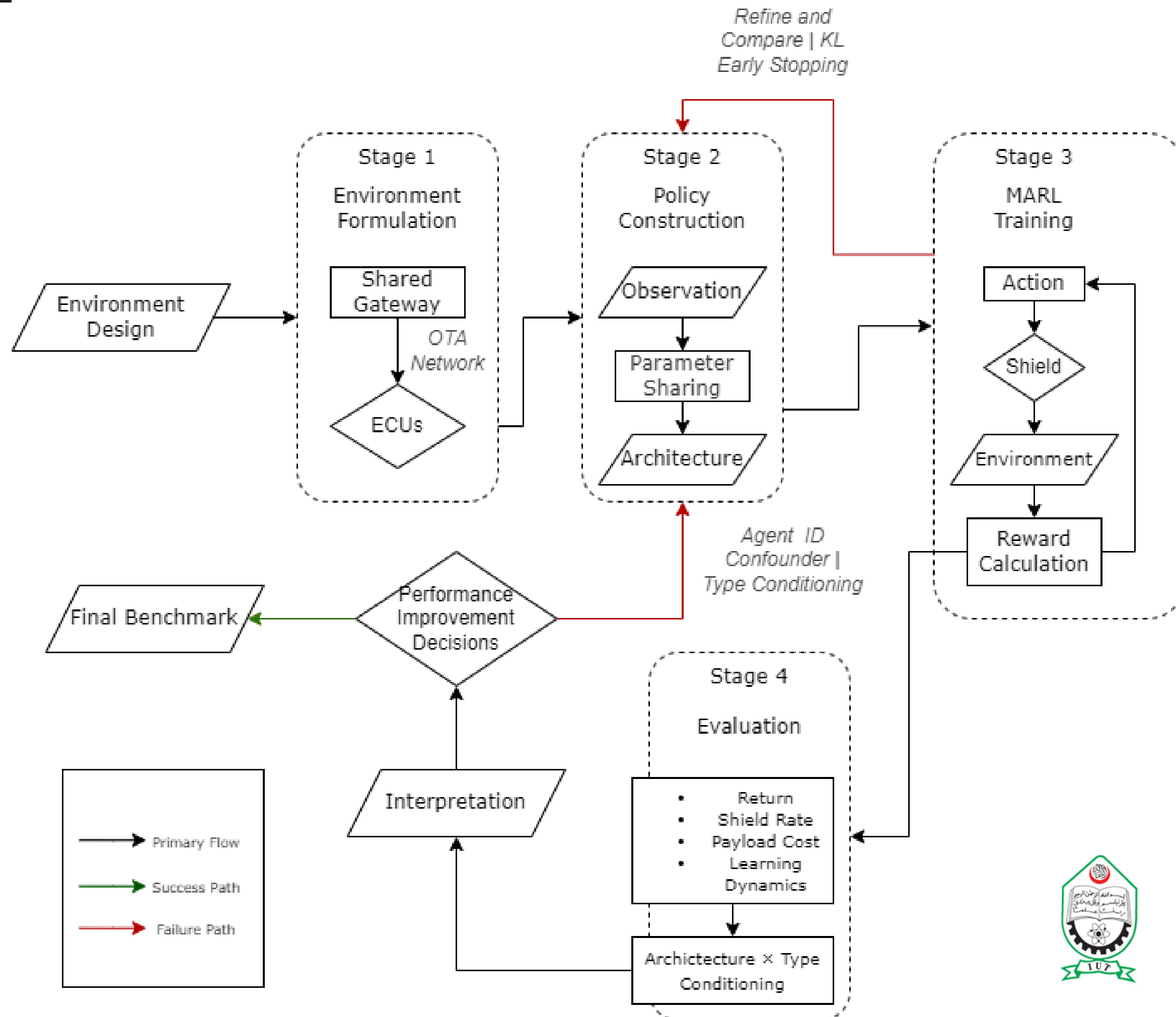
## PROBLEM STATEMENT

“How to effectively coordinate heterogeneous ECUs during shared-channel automotive OTA updates remains unclear, particularly in balancing policy specialization, semantic type information, efficiency, and safety.”

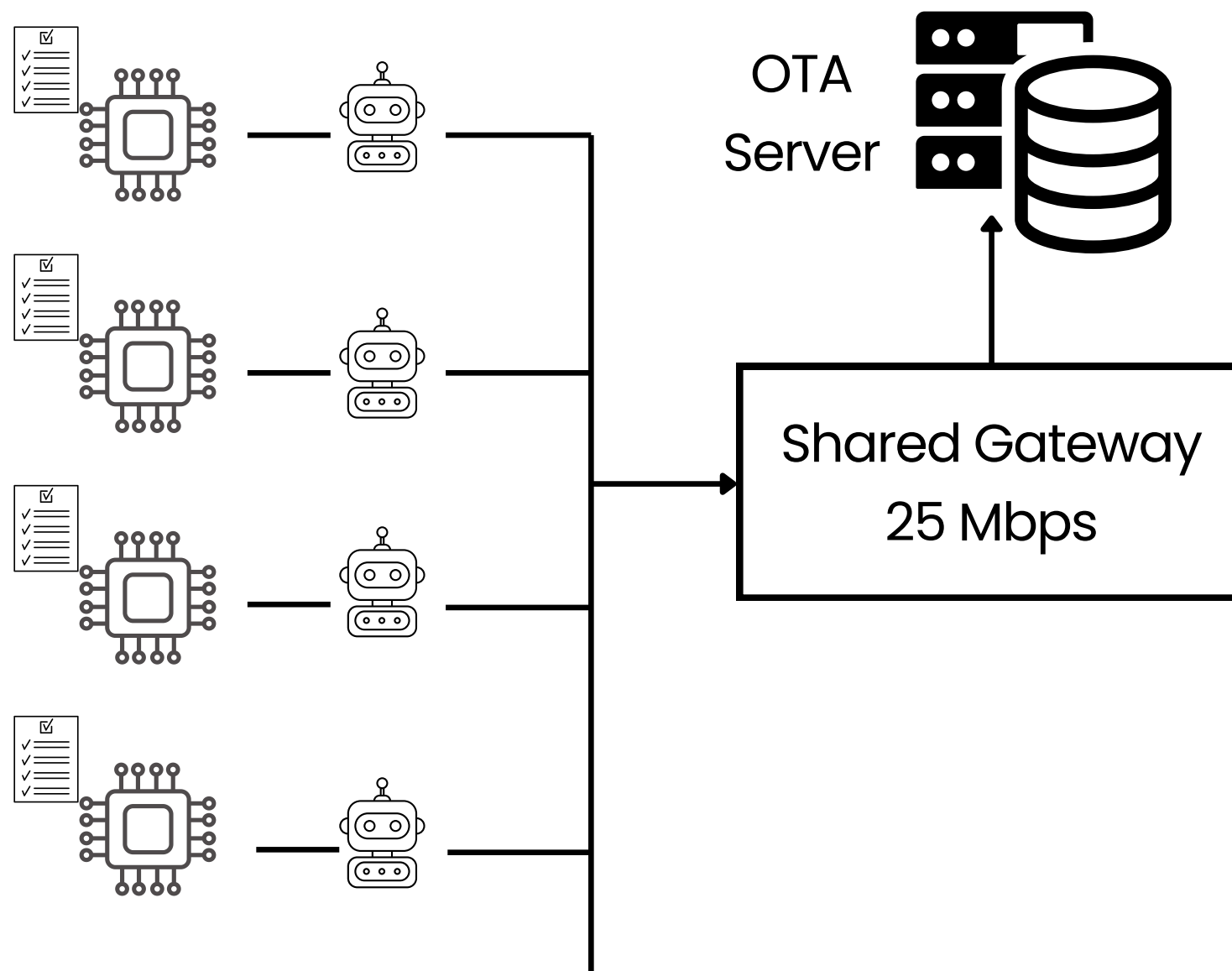


# METHODOLOGY

1. Coupled Channel Environment with PettingZoo + Gymnasium
2. Policy Construction with Stable-Baselines3 + PyTorch
3. MARL Training with safety shielding [10] and shapley credit
4. Evaluation on Mean return, 95% CI, shield rate & payload cost



# ENVIRONMENT FORMULATION



*Note : Automotive OTA architecture aligned with Uptane [09] and UNECE R156.*

## Problem Type : Dec-POMDP [03]

- Partial local observations
- Joint actions determine state transitions
- Cooperative team reward

## Safety & Feasibility

- Memory constraints
- Invalid-action masking
- Deterministic safety shield [10]

## Heterogeneity

- 4 ECU types [07], [08]
- 16 firmware blocks / ECU
- Operation-specific cost preferences [07], [08]

## Channel Model

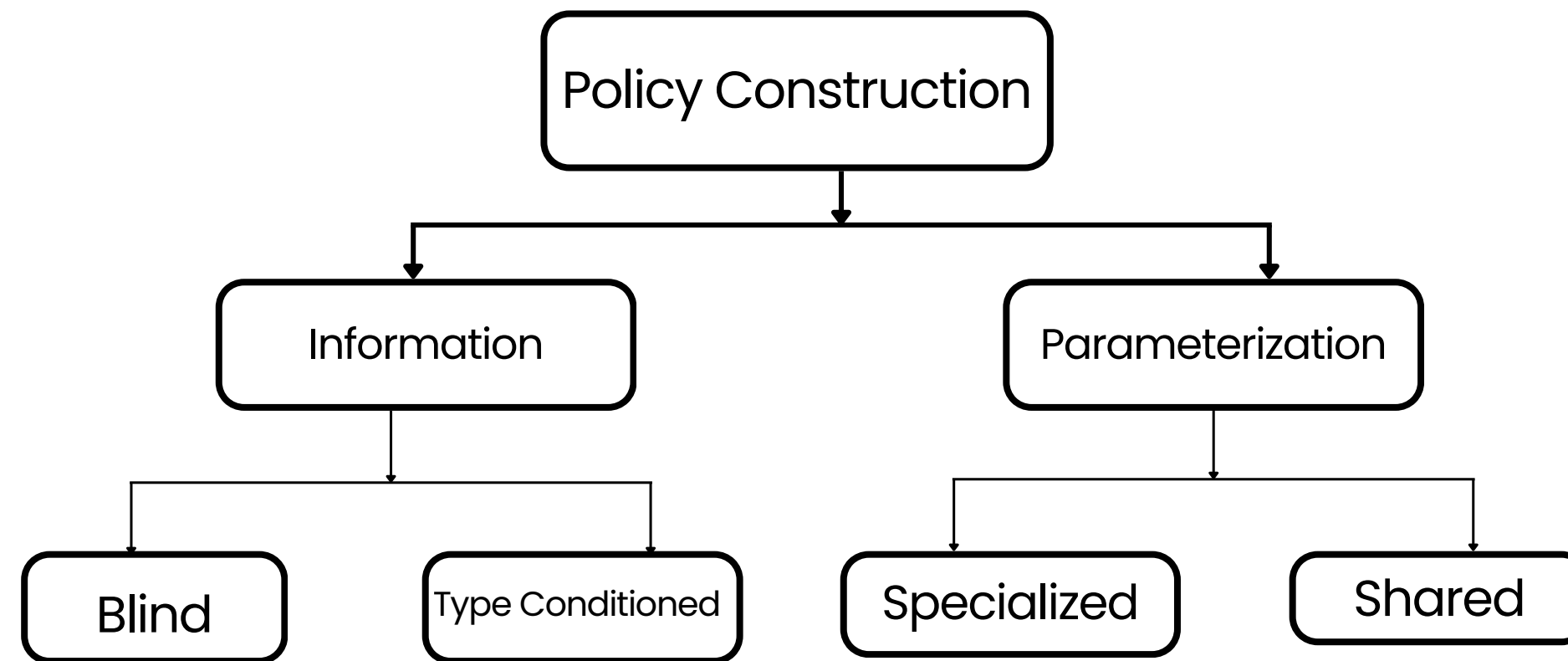
- Shared gateway: 25 Mbps
- Concurrent transmissions divide available capacity
- Stochastic latency + packet loss + burst jitter



Figure: Dec-POMDP Environment Formulation



# POLICY CONSTRUCTION



## Type Conditioning

**Type-Conditioned:** ECU receives semantic type information

**Blind:** ECU type is not explicitly provided

Figure: Policy Construction Taxonomy



# MARL TRAINING

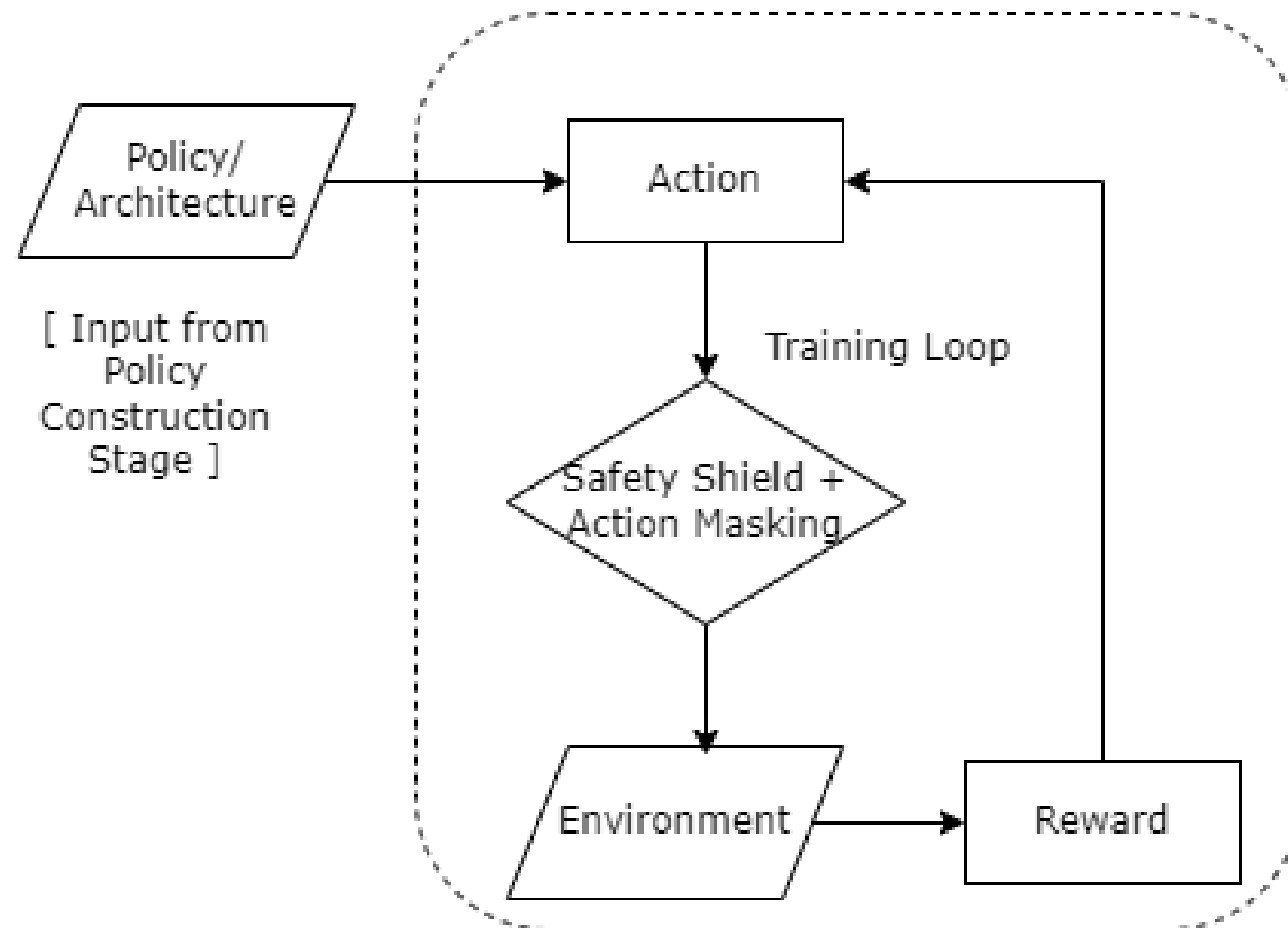


Figure: MARL Training Framework & Optimization Pipeline

Framework : Stable-Baselines3 + PyTorch

## Optimization

- PPO objective [04]
- Adam optimizer
- Learning rate:  $3 \times 10^{-4}$

## Stability

- KL early stopping [04]
- PopArt critic normalization
- Generalized Advantage Estimation

## Co-operative Credit

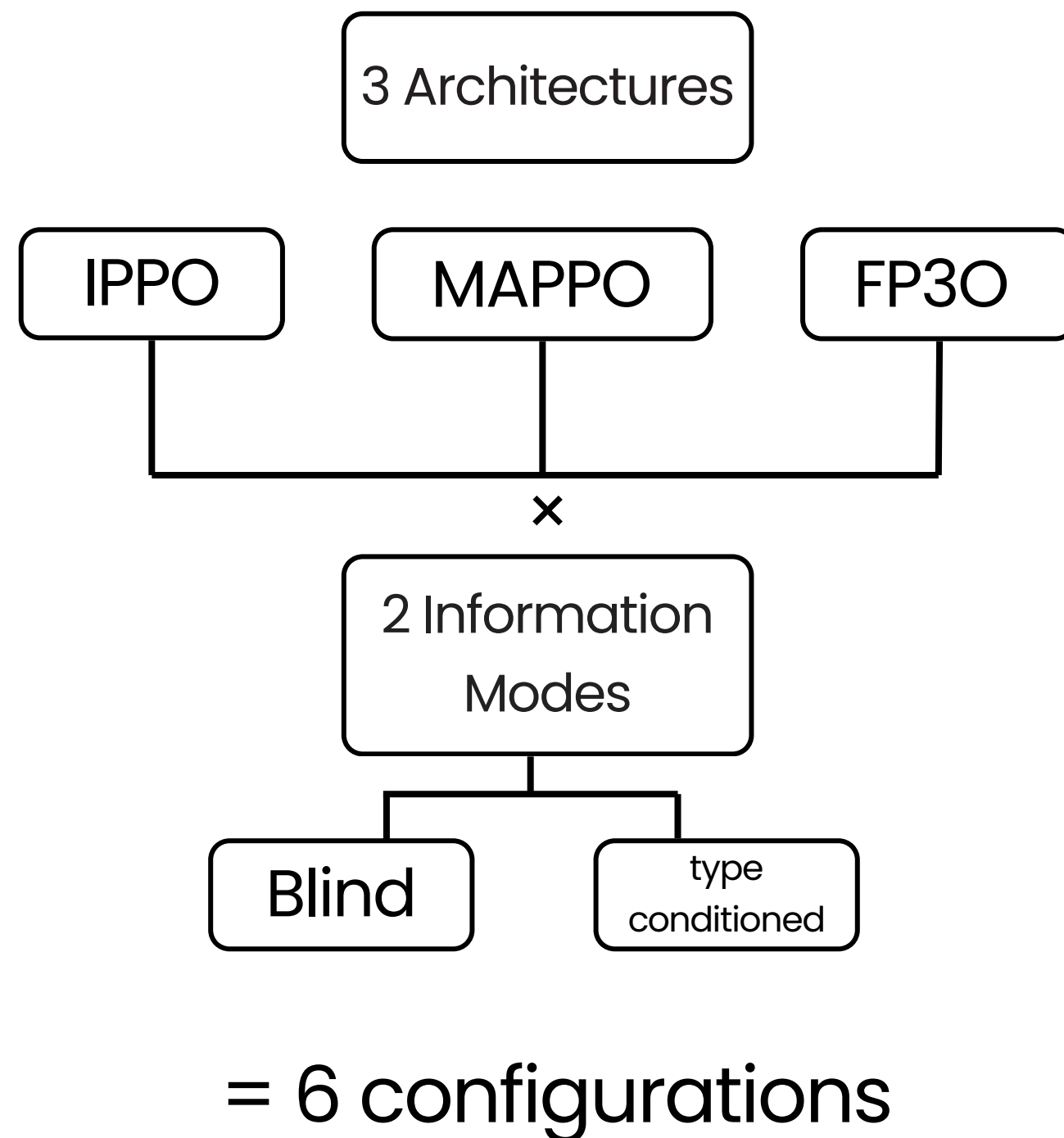
- Shapley-based reward allocation [11]
- Monte Carlo approximation [12]

## Parallelization

- 4 parallel environments
- 256-step rollout
- 1,024 samples / update



# EVALUATION



## Evaluation Metrics

- Mean Episode Return
- Shield Intervention Rate
- Mean Fleet Payload Transmission Cost

## Statistical Analysis

- N independent seeds
- Mean  $\pm$  95% CI



Figure: Experimental Evaluation Setup and Metrics



## REWARD DESIGN

**1. Learning Objective:** Maximize the expected discounted cooperation return

$$J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[ \sum_{t=0}^{T-1} \gamma^t r_t \right] \quad \gamma = 0.99$$

**2. Agent-Level Reward:** +0.5 for a valid, collision-free action

$$r_i^t = \phi_i(v) + r_{\text{prog}} \quad r_{\text{prog}} = +0.5$$

**3. Shapley Credit Calculation:** Compare coalition with ECU  $i$  against the coalition without it [11]

$$\phi_i(v) = \frac{1}{|N|!} \sum_{\pi \in \Pi} [v(S_{\pi}^i \cup \{i\}) - v(S_{\pi}^i)]$$

**4. Coalition Value:**

$$v(S) = \sum_{j \in S \setminus S_{\text{shield}}} \left[ -\alpha(C_{\text{enc}}^j + C_{\text{tx}}^j)m_{\text{type,op}}^j - \beta \text{Overhead}_j + 10 \mathbb{I}(\text{Complete}_j) \right]$$

$$\alpha = 10^{-3}, \beta = 3 \times 10^{-4}$$



# EXPERIMENT SETUP

## Training Horizon

- 4 ECUs
- 16 blocks/ECU
- 25 Mbps
- N seeds
- M Timesteps

- $N = 2 \rightarrow$  Pilot Run
- $N = 5 \rightarrow$  Final Benchmark

- $M = 100K \rightarrow$  Primary Benchmark
- $M = 2M \rightarrow$  Long Horizon Validation

Repository Link :

<https://github.com/saadmansakib47/Multiagent-ReLES-OTA>

## Variables

- Architecture: IPPO | MAPPO | FP3O
- Information: Blind | TC



## EXPERIMENT MATRIX

Debug → Stabilize → Validate → Discover → Pilot → Benchmark

For Benchmark Phase  
Run #(39-44)

|       | Blind | Type-Conditioned |
|-------|-------|------------------|
| IPPO  | ✓     | ✓                |
| MAPPO | ✓     | ✓                |
| FP3O  | ✓     | ✓                |

*Note : Every possible configuration was not exhaustively run. Early results guided the next experimental stage.*



# RESULTS

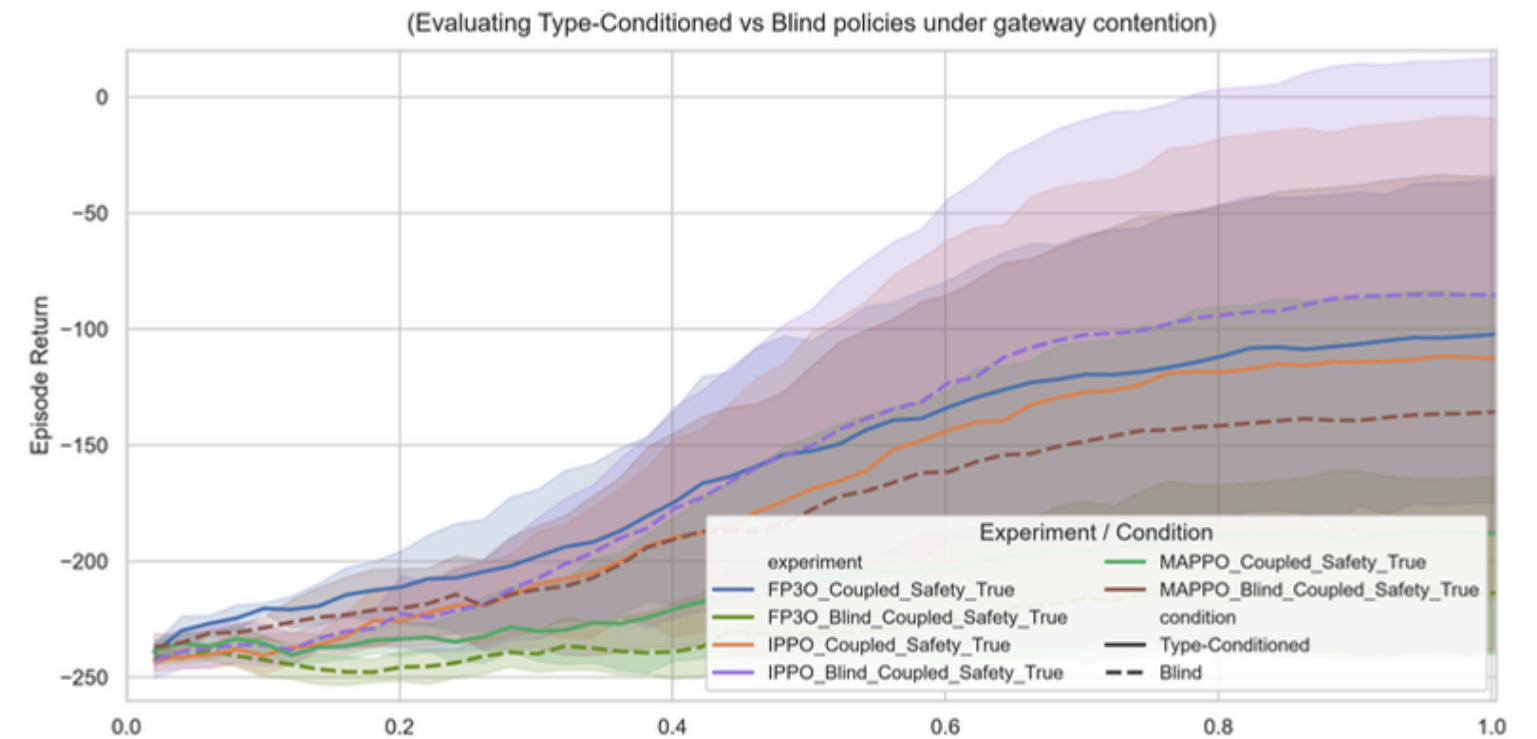
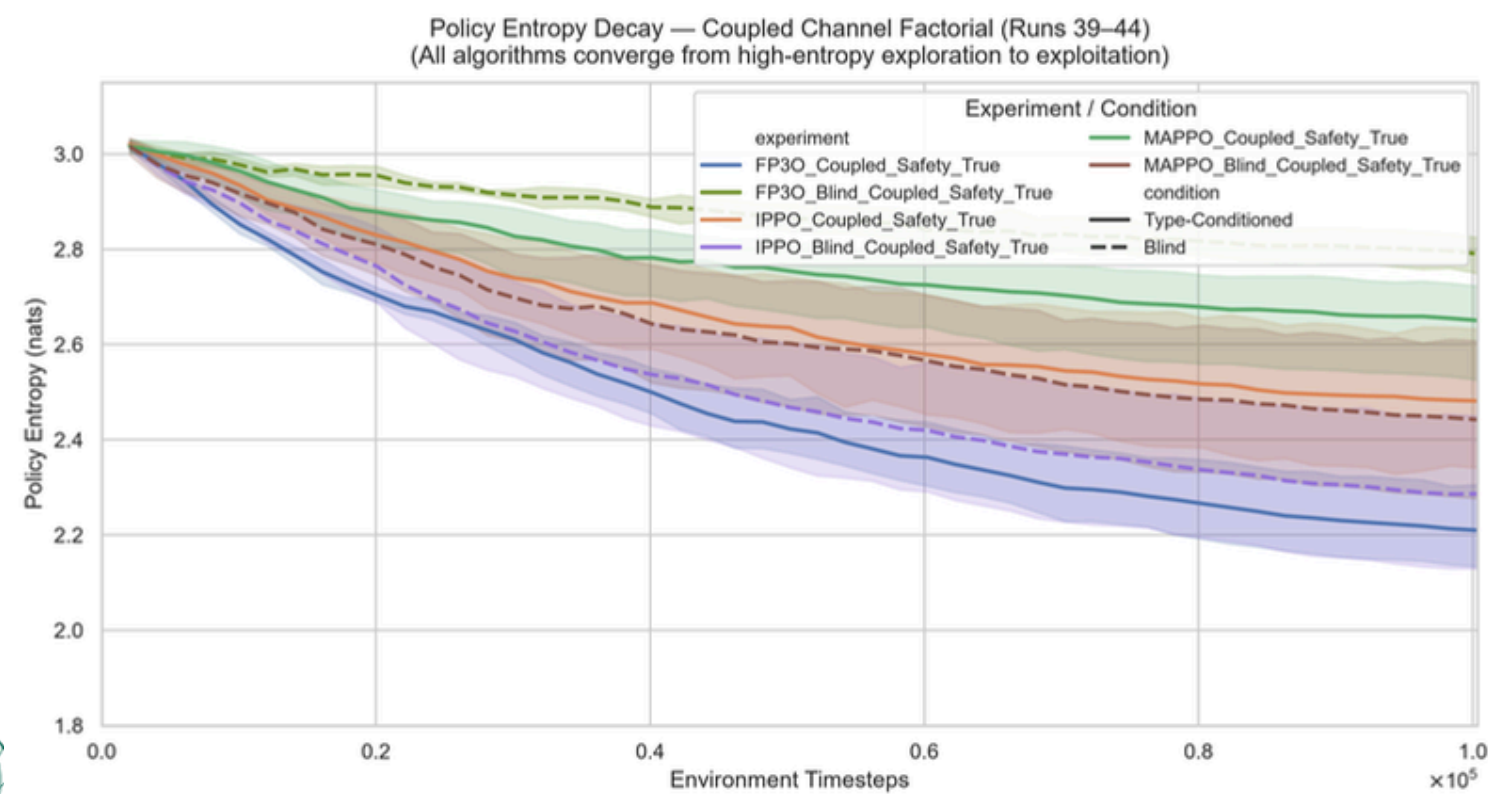
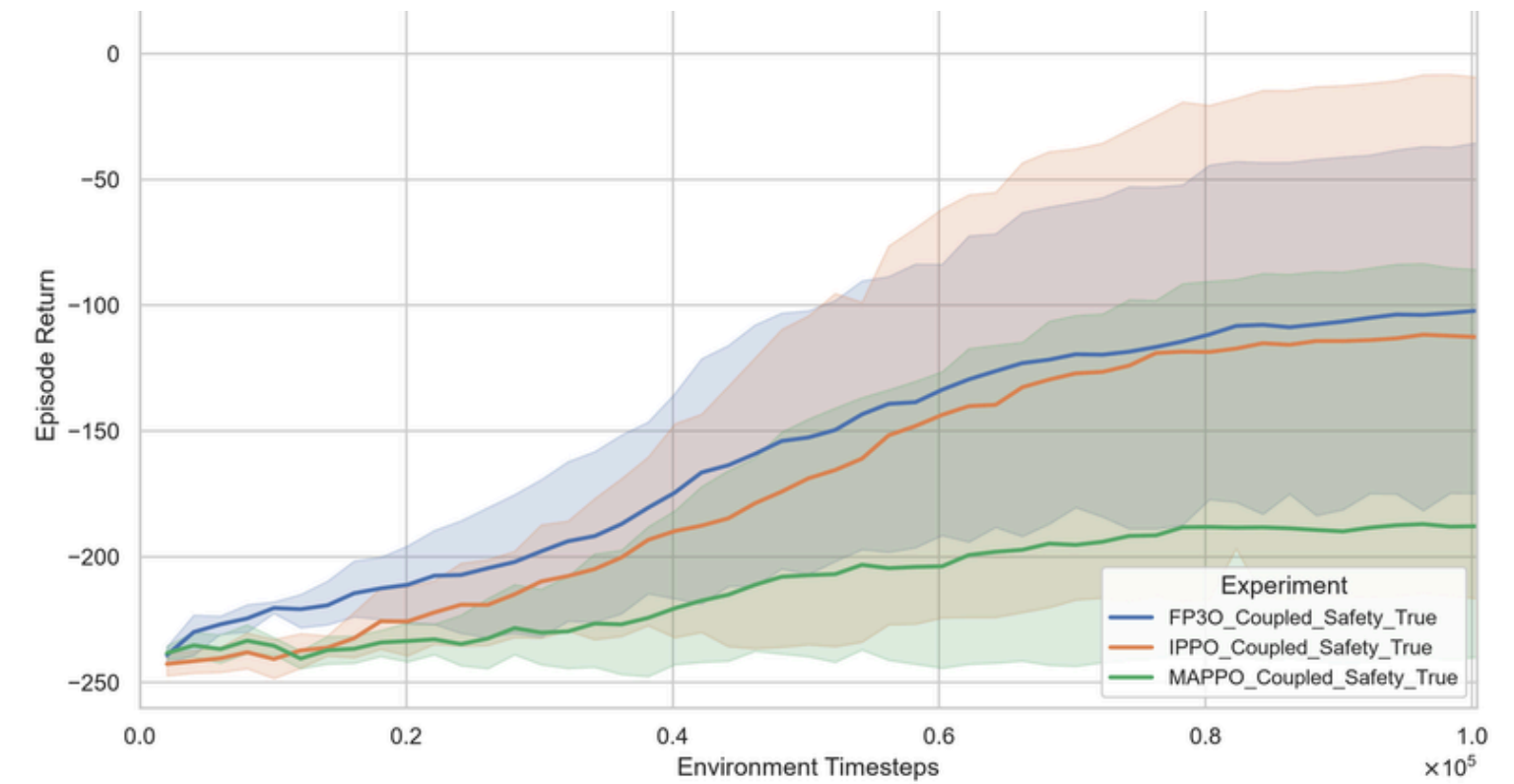
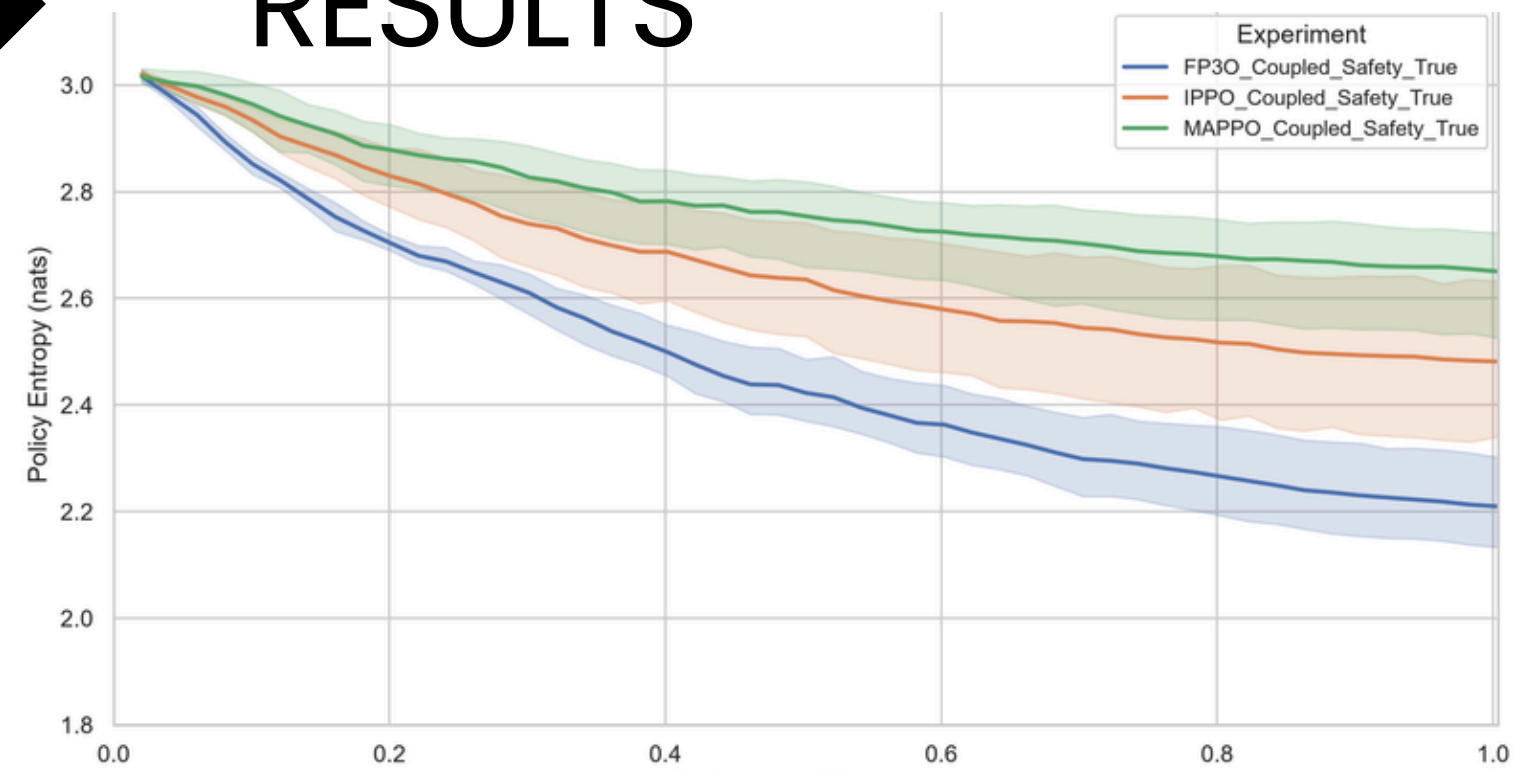


Figure: Entropy Decay and Episode Return Dynamics

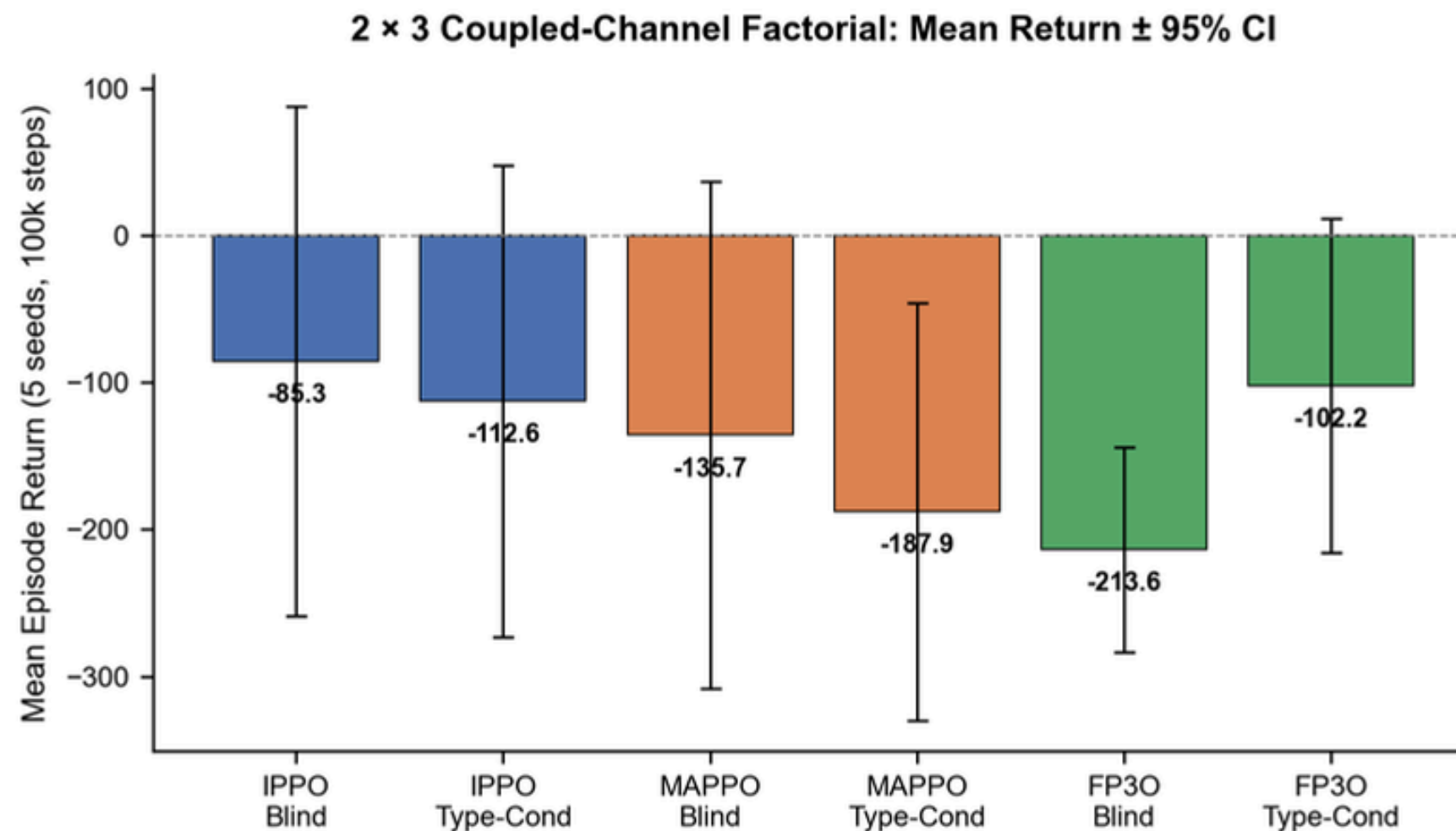


# ARCHITECTURE × CONDITIONING INTERACTION

| Architecture | Conditioning     | Mean Return | 95% CI  | Shield Triggers | Payload (KB) | Interpretation                                                       |
|--------------|------------------|-------------|---------|-----------------|--------------|----------------------------------------------------------------------|
| IPPO         | Blind            | -85.27      | ±173.39 | 32.66%          | 78,239.5     | Best raw return, but considerable shield dependence                  |
| IPPO         | Type-Conditioned | -112.64     | ±160.46 | 32.40%          | 91,953.5     | Conditioning does not improve shared-policy learning                 |
| MAPPO        | Blind            | -135.70     | ±172.42 | 48.41%          | 98,363.8     | Centralized critic alone does not resolve heterogeneity              |
| MAPPO        | Type-Conditioned | -187.88     | ±142.02 | 64.99%          | 118,218.5    | Strongest degradation from conditioning                              |
| FP3O         | Blind            | -213.64     | ±69.75  | 64.68%          | 131,782.3    | Specialized heads without type information are insufficient          |
| FP3O         | Type-Conditioned | -102.20     | ±113.58 | 13.82%          | 106,121.9    | Type information + specialized heads produce the preferred trade-off |



# What Happens with Type Information?



Type-conditioning doesn't help everyone.

- IPPO → Blind is bad, TC is even worse.
- MAPPO → Blind is bad, TC is even worse.
- FP3O → Blind is bad, but TC is better!

Figure: Interaction Effect Between Policy Architecture and Type Information



So, Type information and policy architecture interact.



# THE CONDITIONING EFFECT

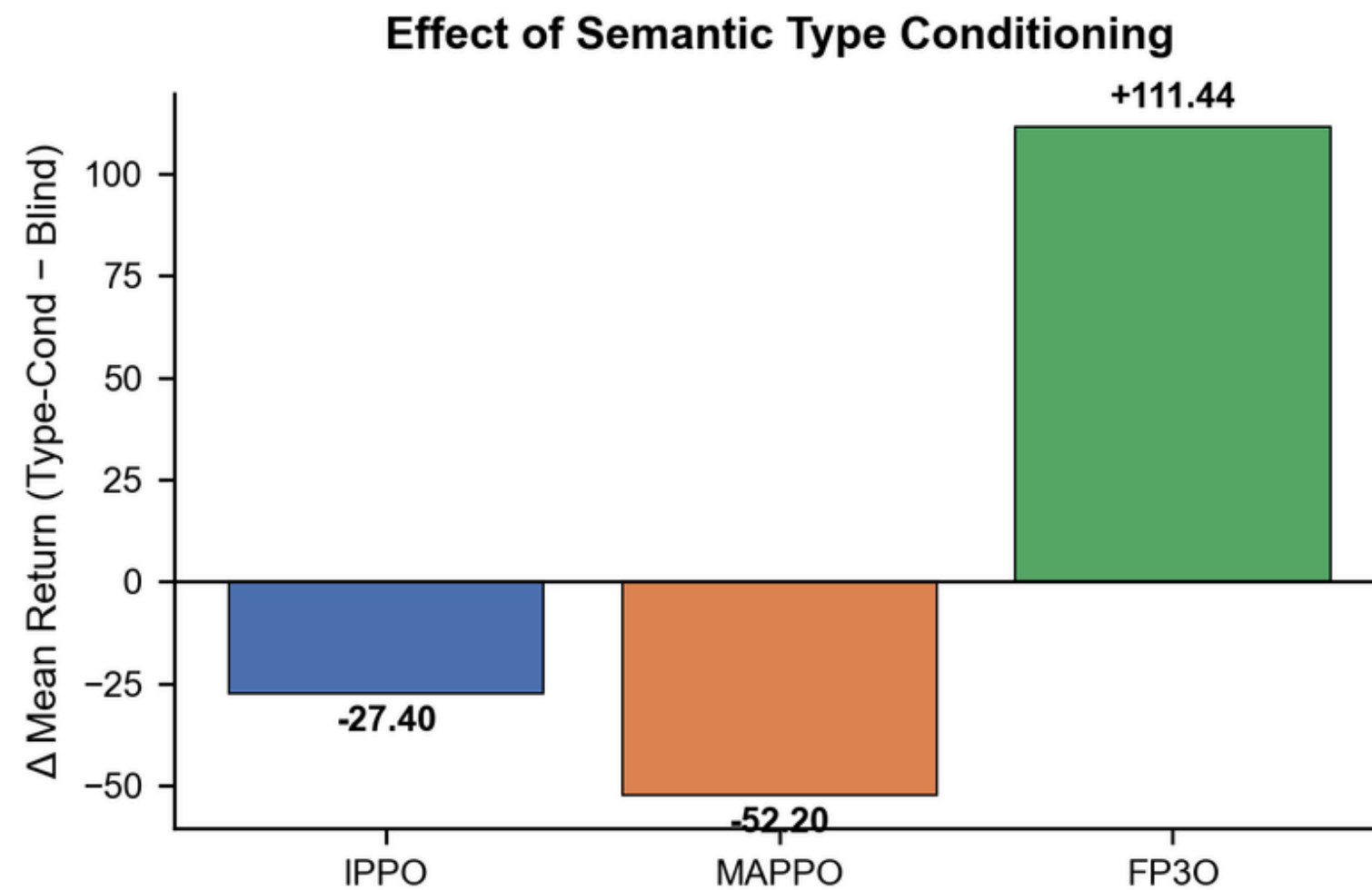
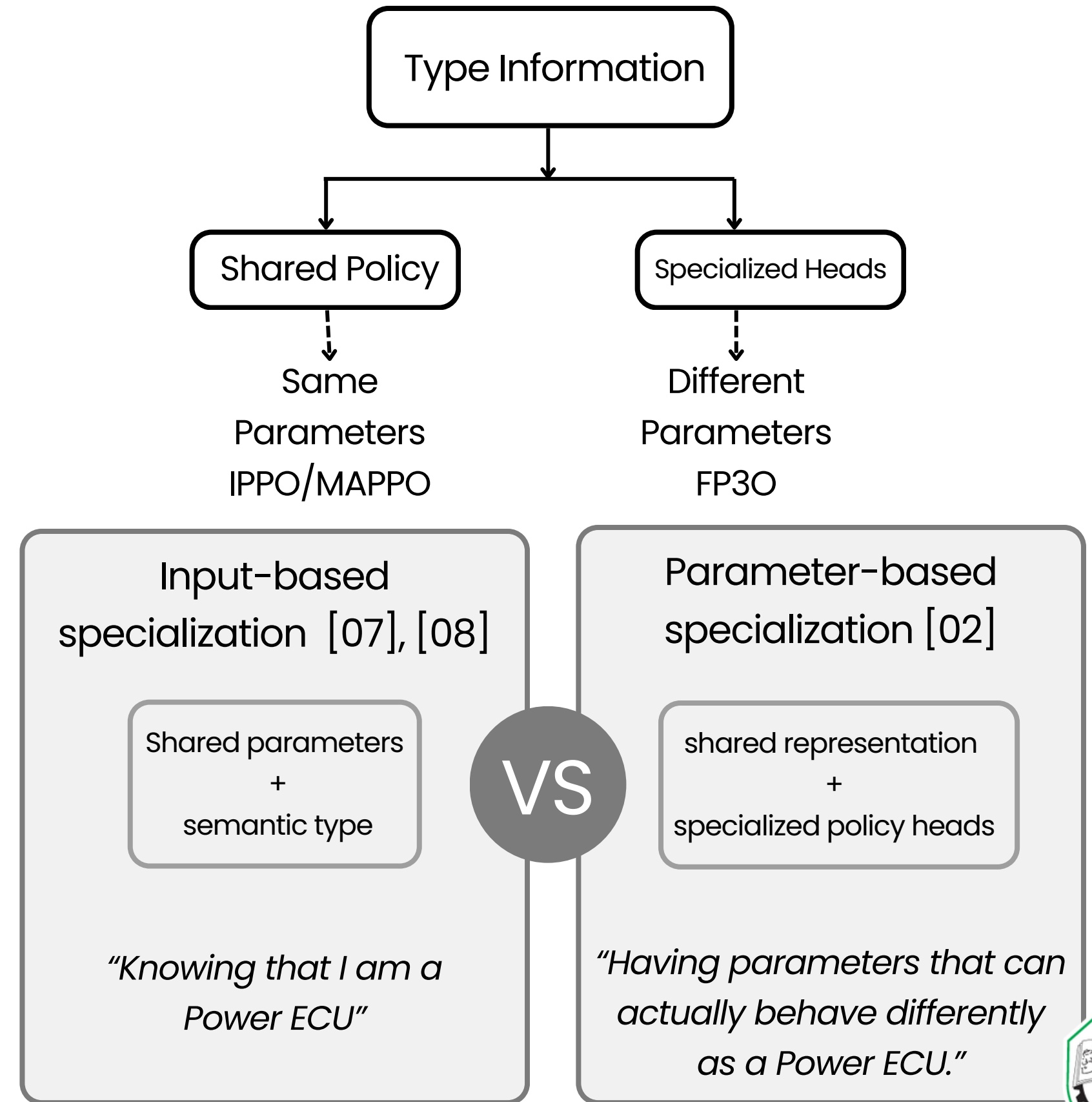


Figure: Comparative Effect of Semantic Type Conditioning Across Architectures



# SAFETY SHIELD TRIGGERING

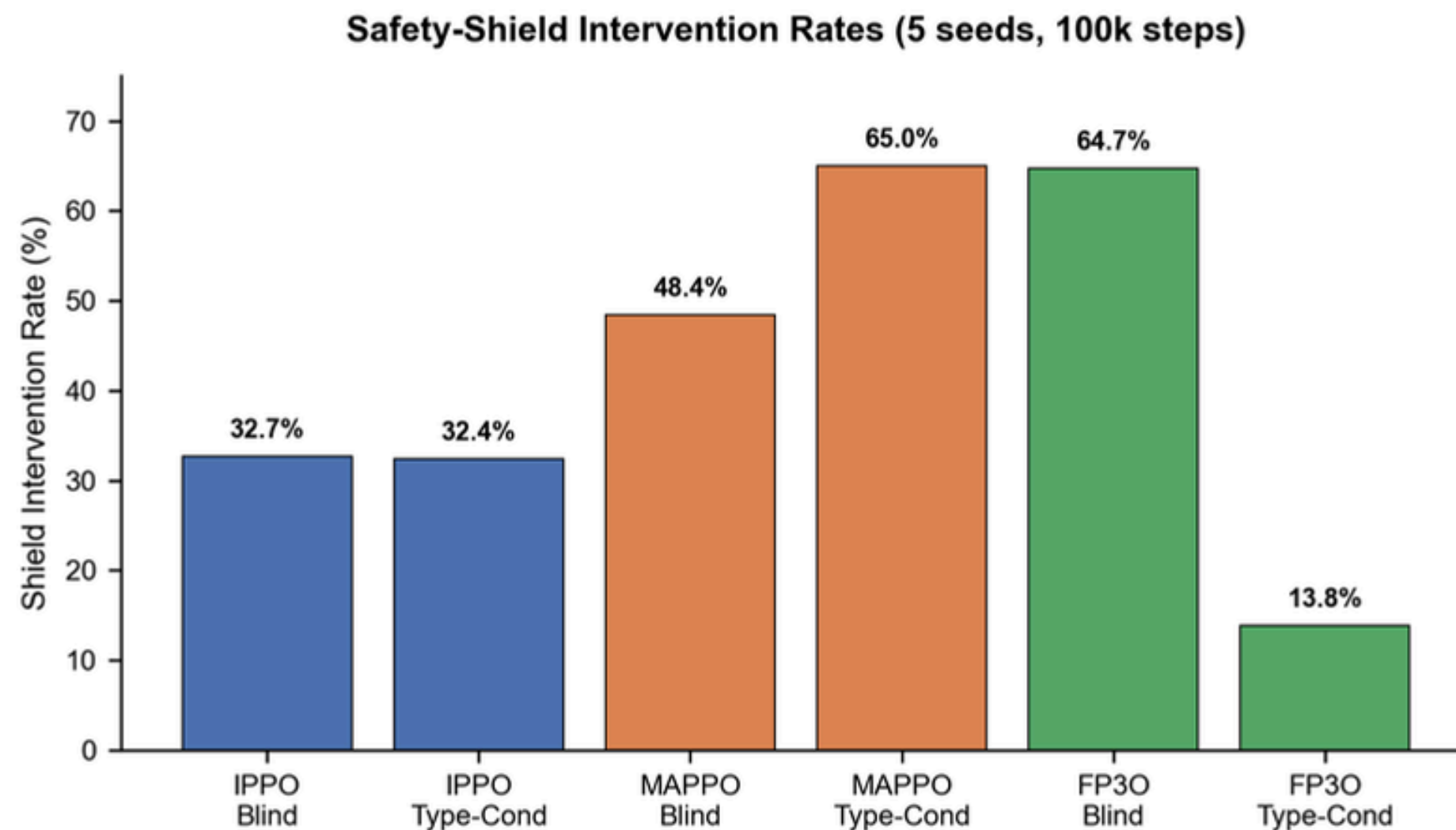


Figure: Safety-Shield Intervention Rates  
Across Configurations

- IPPO Blind achieves the highest return, but remains substantially dependent on the safety shield.
- Type conditioning does not uniformly reduce shield reliance: it decreases slightly for IPPO but increases for MAPPO.
- FP3O shows the strongest conditioning effect: shield activation drops sharply when ECU-type information is provided.
- Lower shield activation suggests less reliance on external safety intervention [10].



## FINDINGS

### Shared Policies

Semantic type information can degrade fully parameter-shared policies under coupled contention.

### Specialized policies

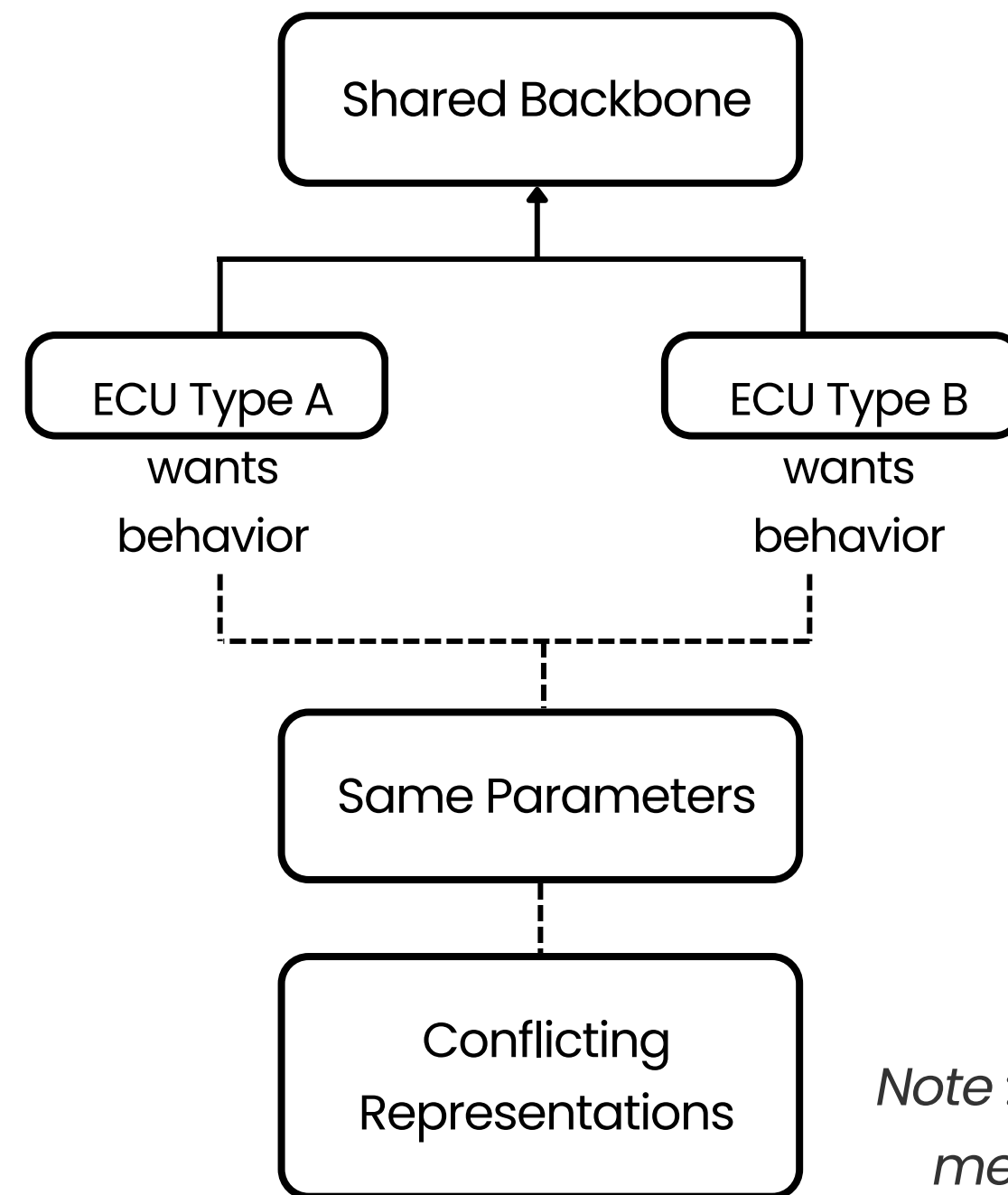
Type conditioning can substantially improve specialized-head FP3O.

### Safety

Numerical return alone can hide shield-dependent behavior.



# FINDINGS (CONT'D)



*Note : Interpretation, not directly measured gradient conflict*

Fully shared policy  
(like IPPO/MAPPO)

Different ECU Roles, but

- Same policy parameters
- Competing representational requirements
- Observed degradation

Specialized heads

Different ECU Roles, plus

- shared features
- separate decision layers
- greater room for specialization



Figure: Conceptual Mechanism of Representation Interference



# CONTRIBUTIONS

| Contribution | Description                                                                      | Status          |
|--------------|----------------------------------------------------------------------------------|-----------------|
| <b>C1</b>    | Heterogeneous MARL formulation for coupled automotive OTA scheduling             | <i>Complete</i> |
| <b>C2</b>    | Shared-channel OTA environment with heterogeneous ECU cost profiles              | <i>Complete</i> |
| <b>C3</b>    | Disentangled-head FP3O architecture for policy specialization                    | <i>Complete</i> |
| <b>C4</b>    | Semantic ECU-type conditioning without agent-identity confounding                | <i>Complete</i> |
| <b>C5</b>    | Controlled comparison of IPPO, MAPPO and FP3O under shared-channel contention    | <i>Complete</i> |
| <b>C6</b>    | Return, safety-shield and fleet payload evaluation across 5 seeds                | <i>Complete</i> |
| <b>C7</b>    | Architecture × conditioning interaction and representation-interference analysis | <i>Complete</i> |
| <b>C8</b>    | Long-horizon validation and reward-saturation analysis                           | <i>Complete</i> |



## LIMITATIONS & FUTURE WORKS

| Direction             | Current Limitation                                                                                | Future Extension                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Heterogeneity         | 4 ECUs                                                                                            | Identify when heterogeneity become large enough to favor specialization [07][08] |
| Safety                | Shield-based evaluation [10]                                                                      | Ablate safety shield and death masking; evaluate learned safety behaviour        |
| Long-Horizon Learning | After $\approx 2M$ timesteps, all configurations approach an empirical reward ceiling near 16.71. | Coordination-sensitive metrics beyond final return                               |
| Realism               | Synthetic Environment                                                                             | Hardware-in-the-loop + real OTA telemetry                                        |



## REFERENCES

- [01] Bhattacharjee et al., ReLES-OTA (2025 *IEEE 3<sup>rd</sup> International Conference on MOST*) – <https://www.weisongshi.org/papers/bhattachajee25-OTA.pdf>
- [02] Feng et al., FP3O (2026 *IEEE TNNLS*) – <https://arxiv.org/abs/2310.05053>
- [03] Yu et al., The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games (*NeurIPS 2022*)– <https://arxiv.org/pdf/2103.01955>
- [04] Christian Schröder de Witt et al., Is Independent Learning All You Need in the StarCraft Multi-Agent Challenge? (*NeurIPS 2020*) – <https://arxiv.org/abs/2011.09533>
- [05] Halder et al., Electric Vehicle Over-the-Air Firmware Updates (*IEEE ICC 2020*) – <https://arxiv.org/pdf/1904.00685>
- [06] Terry et al., Revisiting Parameter Sharing in Multi-Agent Deep RL <https://arxiv.org/pdf/2005.13625>



## REFERENCES

- [07] Zhong et al., Heterogeneous-Agent Reinforcement Learning (*JMLR 2024*) – <https://arxiv.org/pdf/2304.09870>
- [08] Yu et al., Improving Global Parameter-Sharing in Physically Heterogeneous MARL – <https://arxiv.org/pdf/2408.07395>
- [09] Uptane Standard for Design and Implementation, IEEE-ISTO 6100.1.0.0 – <https://uptane.org/papers/ieee-isto-6100.1.0.0.uptane-standard.html>
- [10] Alshiekh et al., Safe Reinforcement Learning via Shielding (*AAAI-18 2018*) – <https://arxiv.org/pdf/1708.08611>
- [11] Shapley, A Value for n-Person Games (Princeton University Press 1953) – <https://www.rand.org/pubs/papers/P295.html>
- [12] Castro et al., Polynomial Calculation of the Shapley Value Based on Sampling (*Computers & Operations Research 2009*) – <https://www.sciencedirect.com/science/article/abs/pii/S0305054808000804>



# APPENDIX A

## Pre-Coupling Learning Curves

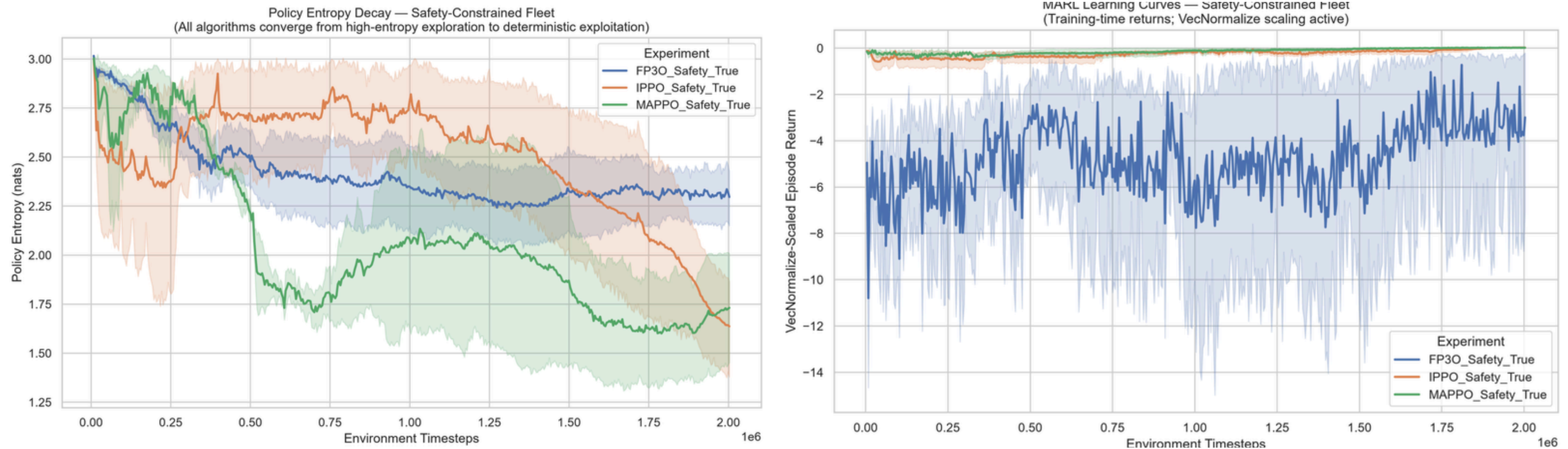


Figure: Pre-Coupling Policy Entropy and Learning Dynamics



# APPENDIX B

## Long-Horizon Reward Saturation

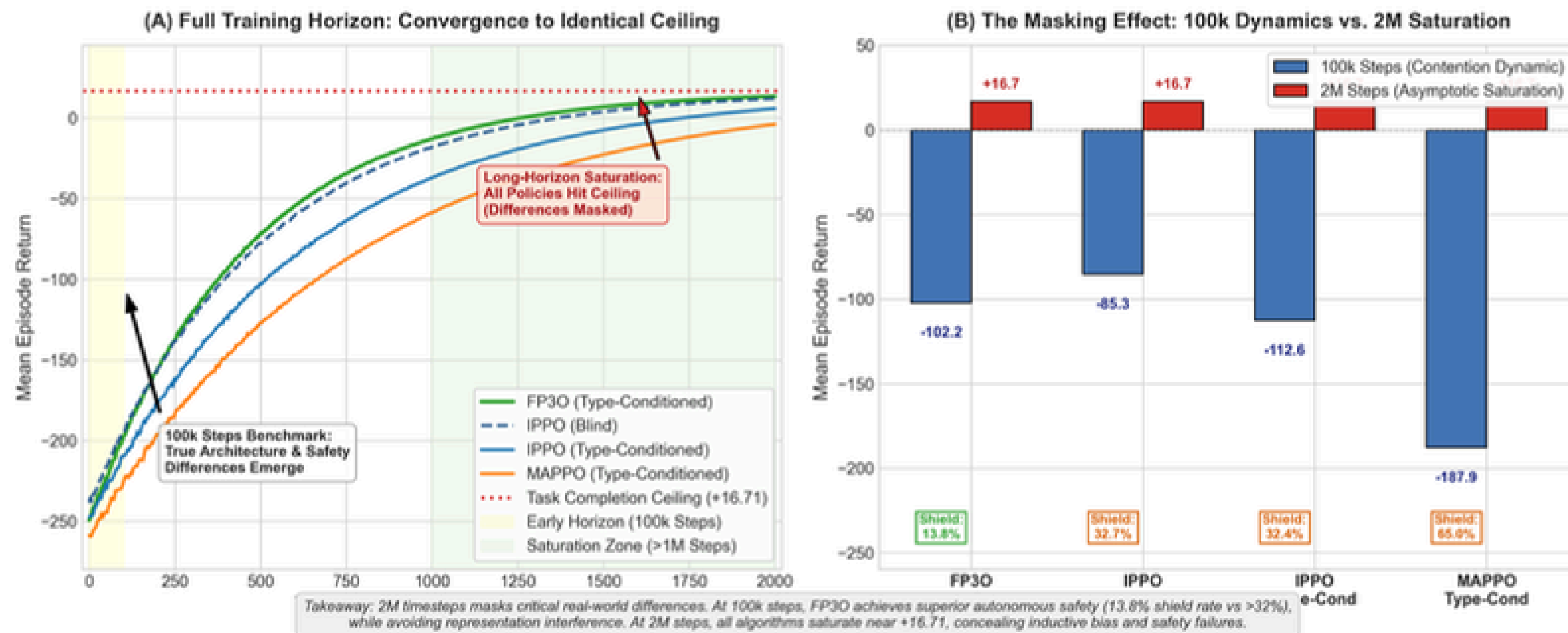
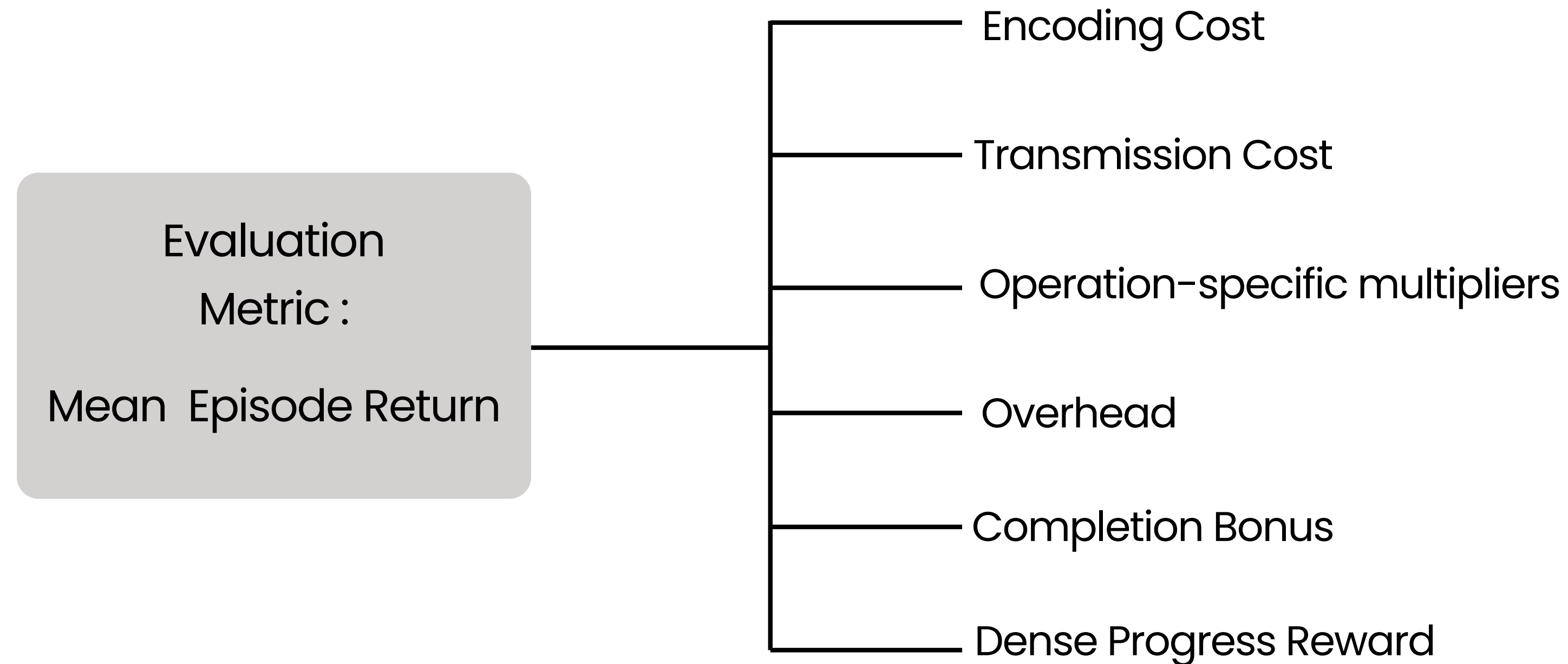


Figure: Long-Horizon Reward Saturation and the Masking Effect



## APPENDIX C

### Construction of Mean Return



# APPENDIX D

## SUPPORTING MECHANISMS

### Type Conditioning

- Each ECU receives its semantic type as input.<sup>[07], [08]</sup>
- This tells the policy what kind of ECU it is.
- It is used to study if type information helps heterogeneous agents.

### Action Masking

- Invalid actions are removed before execution.
- An agent cannot select an impossible block-operation pair.
- Keeps exploration within the feasible action space.

### Safety Shield

- A deterministic shield<sup>[10]</sup> checks memory-safety constraints.
- Unsafe actions are blocked before execution.
- Shield activation is recorded as a behavioral metric.

### Shapley Credit Assignment

- Fleet receives a cooperative outcome.
- Shapley values estimate each agent's contribution.<sup>[11]</sup>
- Helps distribute credit among ECUs.

### KL Early Stopping

- We monitor how much the policy changes after an update.
- Training stops the update early if the change becomes too large.
- This prevents overly aggressive PPO updates.<sup>[04]</sup>

### Stable Training

- GAE reduces noise in advantage estimation.
- PopArt normalizes value targets.
- Multiple parallel environments improve training efficiency.



## APPENDIX E

### Creating Genuine Heterogeneity

Replacing Agent ID with  
semantic type information

Before

After

Agent ID

- 1
- 2
- 3
- 4
- .....

ECU Type

- Engine
- Infotainment
- ADAS
- Body Control
- .....

Identity

Semantics

Type conditioning provides  
semantic information<sup>[07], [08]</sup> without  
giving the policy a unique identity  
shortcut.



# ◆ APPENDIX F

## 44 Training Runs

### Runs 1–12: Pipeline & Reward Debugging

- Oversized reward scale, weak PPO update frequency, invalid-action dominance, safety-shield bug
  - Result: unstable training, extremely poor returns
- Fixes: reward rescaling · smaller rollout · action masking · penalty tuning

### Runs 13–17: Training Stabilization

- Timesteps: 20K → 150K → 1M → 2M (first stable FP3O runs, 2 seeds)
  - Fixed NaN logits / optimizer overflow: logit clamping, target-KL stopping, VecNormalize
  - FP3O positive return, 0% shield activation
- Stable enough for systematic comparison

### Runs 18–25: Baselines & Genuine Heterogeneity

- IPPO & MAPPO added; homogeneous env favored shared policies
- Uniform type-cost scaling failed → operation-specific multipliers revealed conflicting ECU preferences
- Agent-ID confound → replaced with semantic ecu\_type

### Runs 37–44: Final Benchmark + Validation

- Benchmark: 100,353 timesteps × 5 seeds × 6 configurations (identical conditions)
  - Runs 39–44 authoritative; 37–38 = extra 2M coupled validation
- Results used for return, shield rate & payload analysis

### Runs 31–36: Coupled-Channel Pilot

- 100,353 timesteps × 2 seeds, 25 Mbps shared gateway
  - All 6 architecture × information combos; strong differences under contention
  - Type conditioning helped FP3O, hurt shared IPPO/MAPPO; shield activation = key behavioral signal
- Direction identified, scale the decisive comparison

### Runs 26–30: Long-Horizon Scaling

- 2M timesteps × 5 seeds on corrected type-conditioned setup
  - Extended runs approached ≈16.7k reward ceiling
- Long training can hide differences visible earlier!

